

FACULTY OF COMPUTING & INFORMATICS (FCI)

CSN6224 COMPUTER NETWORKS

ASSIGNMENT

Trimester October/November 2025, Term 2530

GROUP 25

A&W

No.	Student ID	Student Name
1	242UC244SU	ALESHA NOOR BINTI MOHAMAD SALIM
2	242UC244PW	ARIANA FALISYA BINTI AIDY @ MOHAMED SHAWAL
3	242UC244LC	AMIRAH BALQIS BINTI AZLAN
4	242UC244T9	WAFIYYAH DIANNAH BINTI HUSSIN

IP Addressing Plan (IPv4 & IPv6)

Subnetting Tables

IPv4 Subnetting Table

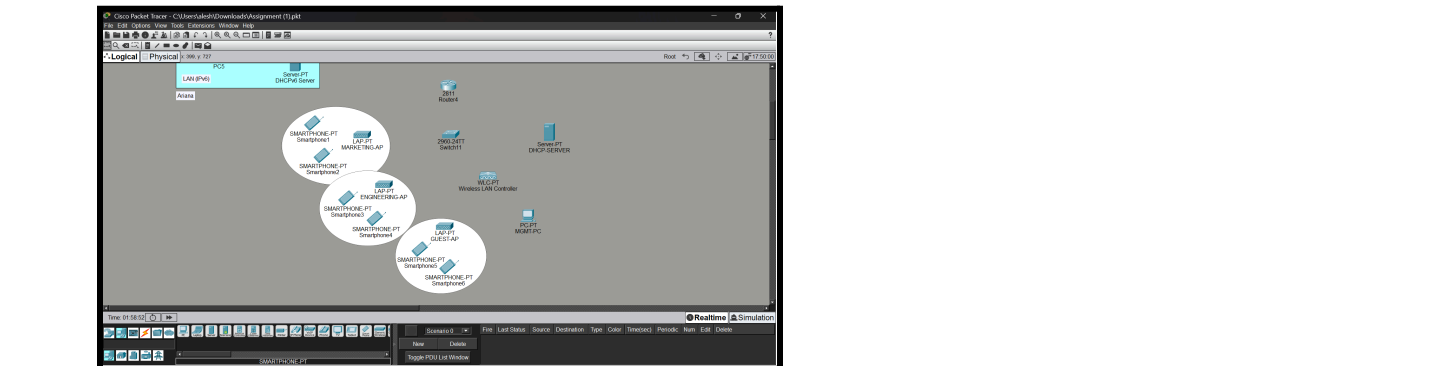
Network	Purpose	Network Address	Prefix	Usable Host Range	Broadcast	Default Gateway	Notes

IPv6 Subnetting Table

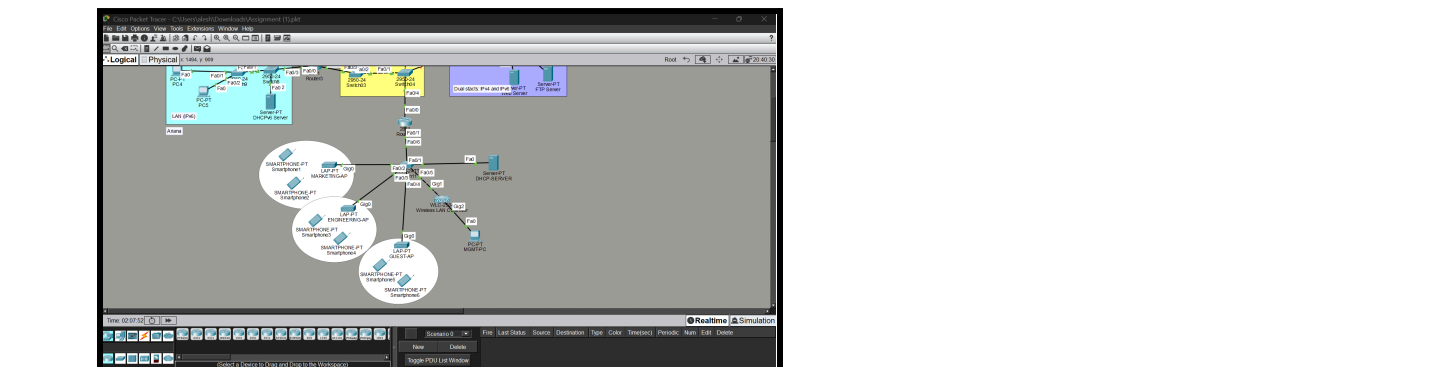
Network	Purpose	IPv6 Prefix	Default Gateway	Usable Host Range	Notes

alesha

1. Place and arrange the devices as shown in the screenshot.



2. Connect all of the devices together.



3.

Test Results (IPv4 & IPv6)

Reflection and Lessons Learned

Name	Reflection	Lessons Learned
Alesha Noor binti Mohamad Salim	I wish I had focused on Sir Khairil's labs more so that I could understand more since I believe it could have cleared up a lot of confusion when I was doing the assignment.	I should've done the assignment earlier on so that I wouldn't be stressing over this assignment when I have three more that I should be worried about.
Ariana Falisya binti Aidy @ Mohamed Shawal		
Amirah Balqis binti Azlan		
Wafiyyah Diannah binti Hussin		

Group Marking Rubric (90%)

Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Group Name:

A&W

Group ID:

Group 25

This document contains the official assessment rubric for your assignment. Marks will be awarded based on the criteria and weightage specified in this rubric, which accounts for **90%** of the **total project marks**. All groups must follow this rubric when completing and presenting their work. The remaining **10%** of the marks will be based on the **Individual Self-Evaluation Form** submitted together with your final report.

Criteria	Weightage (%)	Score 4	Score 3	Score 2	Score 1	Score 0	Rating (Lecturer)
1. VLAN Segmentation & Switch Configuration	15%	VLANs, trunks, STP, redundancy fully correct & stable	Mostly correct with minor errors	VLANs correct but STP/trunks incomplete	Major errors in VLAN or trunk config	Not attempted	
2. IPv4 & IPv6 Addressing Plan (DOB-Based)	10%	Fully correct IPv4/IPv6 addressing with accurate tables	Mostly correct	Some errors or incomplete tables	Incorrect formatting or wrong addressing	Not attempted	
3. Dual-Stack Server Configuration	10%	All services (DNS, Web, Email, FTP) fully functional on IPv4 & IPv6	Most services work; minor issues	Some services functional	Only 1 service working / major issues	No servers configured	
4. Dynamic Routing (IPv4 + IPv6)	10%	Full OSPFv2/OSPFv3 (or RIP) convergence across all routers	Routing works but minor gaps	Partial routing; some unreachable networks	Incorrect routing protocols or major errors	No dynamic routing	
5. Wireless + RADIUS Authentication	10%	All 3 WLANs functional with stable WPA2-Enterprise, DHCP & RADIUS	RADIUS mostly works with minor issues	WLAN OK, RADIUS partially working	Incorrect WLAN/RADIUS setup	Not attempted	
6. End-to-End Connectivity & Simulation Testing	10%	All IPv4/IPv6 ping, services, RADIUS login, simulation verified	Most tests successful	Partial connectivity	Few tests working	No testing conducted	
7. Report Quality & Documentation	10%	Professional, clear, complete, with appendices & evidence	Well-written and complete	Adequate but lacks depth or clarity	Poorly organized	No report submitted	
8. Presentation (Slides + Video)	10%	Excellent clarity, structure, teamwork, technical accuracy	Good delivery with minor gaps	Basic presentation	Poor clarity or missing parts	No presentation submitted	
9. Bonus: Enhancements (SSH, ACLs, Redundancy, Syslog/SNMP)	5%	3+ enhancements fully functional	Multiple enhancements with minor issues	1–2 enhancements partially working	Attempted but incorrect	No enhancements	

Weekly Project Guidelines / Checklist (Week 5 – 13)

Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Group Name: A&W Group ID: Group 25

These guidelines serve as a weekly reminder of the suggested tasks for your group. They are intended to help you plan and complete your project effectively. You are not required to follow them strictly, and your group may adjust the schedule as needed based on progress and workload.

Week	Task Title	Task Item	Checklist
Week 5	Group Formation	Form group & confirm member list	✓
		Set up communication channel	✓
		Assign roles (Leader, Technical Lead, Documentation Lead, Presentation Lead, QA)	✓
		Conduct first meeting	✓
Week 6	Requirements Analysis	Study assignment requirements	✓
		Develop project scenario	✓
		Divide tasks among group members	✓
		Prepare high-level network design	✓
		Lecturer check-in: confirm scope and design idea	
Week 7	Initial Network Design	Draft topology in Packet Tracer	
		Start IP addressing plan	
		Identify required protocols	
		Lecturer check-in: verify topology and addressing draft	
Week 8	Requirements Mapping	Finalize topology	
		Begin basic configurations	
		Lecturer check-in: confirm all requirements covered	
Week 9	Core Implementation	Configure routing, switching, DHCP/NAT/ACL	
		Conduct basic connectivity tests	
		Lecturer check-in: demonstrate partial working network	
Week 10	Troubleshooting & Enhancement	Fix issues and refine configurations	
		Add enhancements (if allowed)	
		Lecturer check-in: verify functionality	
Week 11	Documentation & Review	Finalize topology, tables, configuration summaries, test results	
		QA check	
		Lecturer check-in: review near-final work	
Week 12	Final Report & Video	Finalize written report	
		Finalize presentation slides	
		Record project video presentation	
		Upload video to YouTube (Unlisted)	
Week 13	Submission to eBwise	Submit Packet Tracer file (.pkt)	
		Submit PDF report	
		Submit presentation slides (.pptx or other format)	

Group Name: A&W

NO.	TASK DESCRIPTION	MEMBER-IN -CHARGE	DATE ASSIGNED	DATE COMPLETED	REMARK
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

**Note: You may edit the format of this page.*

Prepared by:

.....
Group Leader

Name: _____

ID : _____

Individual Self-Evaluation Form (10%) Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name: ALESHA NOOR BINTI MOHAMAD SALIM
Group Name: A&W

Student ID: 242UC244SU
Group ID: Group 25

Each group member must complete and sign this form to verify the accuracy of their self-evaluation.
Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**.
The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

Criteria	Weightage (%)	4 – Excellent	3 – Good	2 – Fair	1 – Poor	0 – None	Your Rating (Number)
Attendance	2	Attended all meetings except excused official tasks (with proof).	Missed 1 meeting with acceptable reason.	Missed 2–3 meetings; explanation acceptable but weak.	Frequently absent; weak or unclear reasons.	No attendance / no explanation.	
Task Completion	3	Completed all tasks on time with clear evidence.	Completed most tasks; minor delays.	Completed some tasks; multiple delays.	Very minimal task completion.	No completed tasks.	
Communication	1.5	Always responsive; active in chat & group activities.	Mostly responsive; occasional delay.	Sometimes responds; inconsistent.	Rarely responds; slow communication.	No communication at all.	
Contribution	2	Actively contributes ideas and solutions in discussions.	Regular contributions; useful input.	Occasional contributions; basic input.	Rarely contributes; passive.	No contribution.	
Initiative	1.5	Proactively initiates tasks and helps others consistently.	Often shows initiative and helps when needed.	Sometimes helps; limited initiative.	Rarely initiates; only helps when asked.	No initiative or avoids helping.	

By signing below, you confirm that all information provided in this form is true and accurate.

Declaration

☐

I hereby confirm that the above self-evaluation is accurate.

.....
Student Name:

Student ID:

Date:

Individual Self-Evaluation Form (10%) Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name:	ARIANA FALISYA BINTI AIDY @ MOHAMED SHAWAL	Student ID:	242UC244PW
Group Name:	A&W	Group ID:	Group 25

Each group member must complete and sign this form to verify the accuracy of their self-evaluation.
Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**.
The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

Criteria	Weightage (%)	4 – Excellent	3 – Good	2 – Fair	1 – Poor	0 – None	Your Rating (Number)
Attendance	2	Attended all meetings except excused official tasks (with proof).	Missed 1 meeting with acceptable reason.	Missed 2–3 meetings; explanation acceptable but weak.	Frequently absent; weak or unclear reasons.	No attendance / no explanation.	4
Task Completion	3	Completed all tasks on time with clear evidence.	Completed most tasks; minor delays.	Completed some tasks; multiple delays.	Very minimal task completion.	No completed tasks.	4
Communication	1.5	Always responsive; active in chat & group activities.	Mostly responsive; occasional delay.	Sometimes responds; inconsistent.	Rarely responds; slow communication.	No communication at all.	4
Contribution	2	Actively contributes ideas and solutions in discussions.	Regular contributions; useful input.	Occasional contributions; basic input.	Rarely contributes; passive.	No contribution.	4
Initiative	1.5	Proactively initiates tasks and helps others consistently.	Often shows initiative and helps when needed.	Sometimes helps; limited initiative.	Rarely initiates; only helps when asked.	No initiative or avoids helping.	4

By signing below, you confirm that all information provided in this form is true and accurate.

Declaration



I hereby confirm that the above self-evaluation is accurate.



.....
Student Name: ARIANA FALISYA BINTI AIDY @ MOHAMED SHAWAL

Student ID: 242UC244PW

Date: 29/1/2026

Individual Self-Evaluation Form (10%) Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name: AMIRAH BALQIS BINTI AZLAN
Group Name: A&W

Student ID: 242UC244LC
Group ID: Group 25

Each group member must complete and sign this form to verify the accuracy of their self-evaluation.
Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**.
The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

Criteria	Weightage (%)	4 – Excellent	3 – Good	2 – Fair	1 – Poor	0 – None	Your Rating (Number)
Attendance	2	Attended all meetings except excused official tasks (with proof).	Missed 1 meeting with acceptable reason.	Missed 2–3 meetings; explanation acceptable but weak.	Frequently absent; weak or unclear reasons.	No attendance / no explanation.	
Task Completion	3	Completed all tasks on time with clear evidence.	Completed most tasks; minor delays.	Completed some tasks; multiple delays.	Very minimal task completion.	No completed tasks.	
Communication	1.5	Always responsive; active in chat & group activities.	Mostly responsive; occasional delay.	Sometimes responds; inconsistent.	Rarely responds; slow communication.	No communication at all.	
Contribution	2	Actively contributes ideas and solutions in discussions.	Regular contributions; useful input.	Occasional contributions; basic input.	Rarely contributes; passive.	No contribution.	
Initiative	1.5	Proactively initiates tasks and helps others consistently.	Often shows initiative and helps when needed.	Sometimes helps; limited initiative.	Rarely initiates; only helps when asked.	No initiative or avoids helping.	

By signing below, you confirm that all information provided in this form is true and accurate.

Declaration

☐

I hereby confirm that the above self-evaluation is accurate.

.....
Student Name:

Student ID:

Date:

Individual Self-Evaluation Form (10%) Assignment

CSN6224 Communication Networks
Trimester October/November 2025 Term 2530

Student Name: WAFIYYAH DIANNAH BINTI HUSSIN
Group Name: A&W

Student ID: 242UC244T9
Group ID: Group 25

Each group member must complete and sign this form to verify the accuracy of their self-evaluation.
Please **attach** the completed form together with your group report during submission.

If you have **concerns** or **disagree** with any part of a group member's self-evaluation, please **email** the lecturer **privately**.
The lecturer will review the matter, and appropriate action will be taken if the concern is substantiated.

Criteria	Weightage (%)	4 – Excellent	3 – Good	2 – Fair	1 – Poor	0 – None	Your Rating (Number)
Attendance	2	Attended all meetings except excused official tasks (with proof).	Missed 1 meeting with acceptable reason.	Missed 2–3 meetings; explanation acceptable but weak.	Frequently absent; weak or unclear reasons.	No attendance / no explanation.	4
Task Completion	3	Completed all tasks on time with clear evidence.	Completed most tasks; minor delays.	Completed some tasks; multiple delays.	Very minimal task completion.	No completed tasks.	4
Communication	1.5	Always responsive; active in chat & group activities.	Mostly responsive; occasional delay.	Sometimes responds; inconsistent.	Rarely responds; slow communication.	No communication at all.	4
Contribution	2	Actively contributes ideas and solutions in discussions.	Regular contributions; useful input.	Occasional contributions; basic input.	Rarely contributes; passive.	No contribution.	4
Initiative	1.5	Proactively initiates tasks and helps others consistently.	Often shows initiative and helps when needed.	Sometimes helps; limited initiative.	Rarely initiates; only helps when asked.	No initiative or avoids helping.	4

By signing below, you confirm that all information provided in this form is true and accurate.

Declaration



I hereby confirm that the above self-evaluation is accurate.



.....
Student Name:

Student ID:

Date: